Intelligence Academy

Computer Graphics With Mathematics and Python

Lecture 04: Linear Algebra for Transformations

Mejbah Ahammad

Overview and Mental Model

From Scene to Pixels

A rendering pipeline maps a *scene* (geometry, materials, lights, camera) to *pixels*. Each stage transforms data: spaces (model/world/view/clip), topology (triangles), and samples (fragments) before writing to the framebuffer.

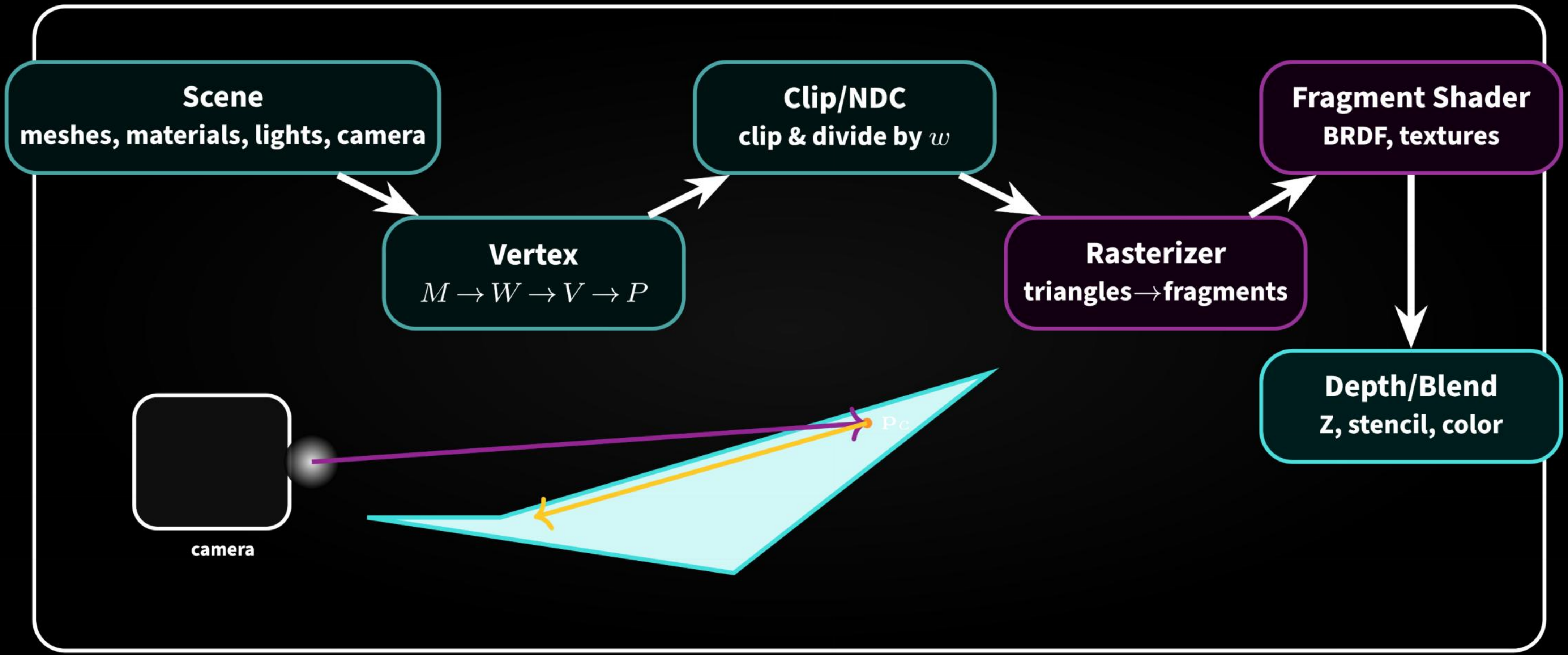

Figure 1: Pipeline overview. Scene data flow through geometry stages, then the rasterizer, then per-fragment shading and output (depth/blend).

Mathematics (core transforms).

$$\mathbf{x}_{clip} = P\,V\,W\,M\,\mathbf{x}_{model}, \qquad \mathbf{x}_{ndc} = \left(\frac{x_{clip}}{w_{clip}}, \frac{y_{clip}}{w_{clip}}, \frac{z_{clip}}{w_{clip}}\right).$$

Screen mapping (viewport with width W, height H):

$$x_{scr} = \frac{W}{2}(x_{ndc}+1), \qquad y_{scr} = \frac{H}{2}(1-y_{ndc}) \quad \text{(top-left origin).}$$

Worked Example (clip $\to$ NDC $\to$ screen).

Let $W{=}1280$, $H{=}720$, and $\mathbf{x}_{clip} = (2,-1,3,4)$. Then

$$\mathbf{x}_{ndc} = \left(\tfrac{2}{4}, \tfrac{-1}{4}, \tfrac{3}{4}\right) = (0.5, -0.25, 0.75).$$

Hence $x_{scr} = 1280 \cdot \frac{0.5+1}{2} = 960, \quad y_{scr} = 720 \cdot \frac{1-(-0.25)}{2} = 720 \cdot 0.625 = 450.$

- Clipping before divide: Always clip in clip space to avoid slicing artifacts.
- Homogeneous role: w encodes perspective; division maps frustum to NDC cube.
- Viewport origin: APIs differ (GL vs DX). Document your screen-space convention.

Fixed-Function vs. Programmable Stages

Idea. Some stages are *fixed-function* (hardware-defined: rasterization, depth test, blending). Others are *programmable* shaders (vertex/fragment, and optional tessellation/geometry/mesh, compute).

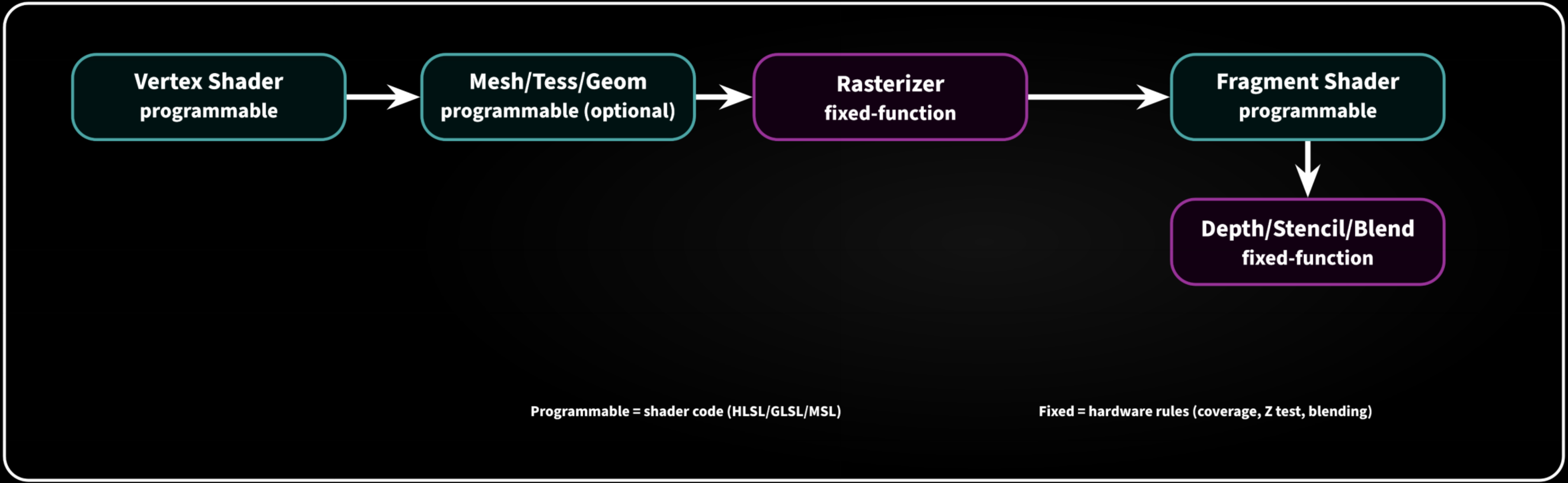

Figure 2: Programmable vs. fixed-function. Shaders do math and lookups; fixed stages enforce coverage, depth/stencil, and blending.

Raster edge functions and barycentrics.

Given triangle vertices $A(x_a, y_a)$, $B(x_b, y_b)$, $C(x_c, y_c)$, define edge functions

$$E_{AB}(x,y) = (y_a - y_b)x + (x_b - x_a)y + x_a y_b - x_b y_a,$$

and cyclic permutations. A sample (x, y) is inside (with top-left rule) iff

$$E_{AB} \geq 0, \quad E_{BC} \geq 0, \quad E_{CA} \geq 0.$$

Barycentrics (unnormalized): $w_a = E_{BC}(x,y),\ w_b = E_{CA}(x,y),\ w_c = E_{AB}(x,y)$, then

$$\lambda_i = \frac{w_i}{w_a + w_b + w_c}, \qquad \mathbf{v}(x,y) = \lambda_a\, \mathbf{v}_a + \lambda_b\, \mathbf{v}_b + \lambda_c\, \mathbf{v}_c.$$

pixel coverage and interpolation.

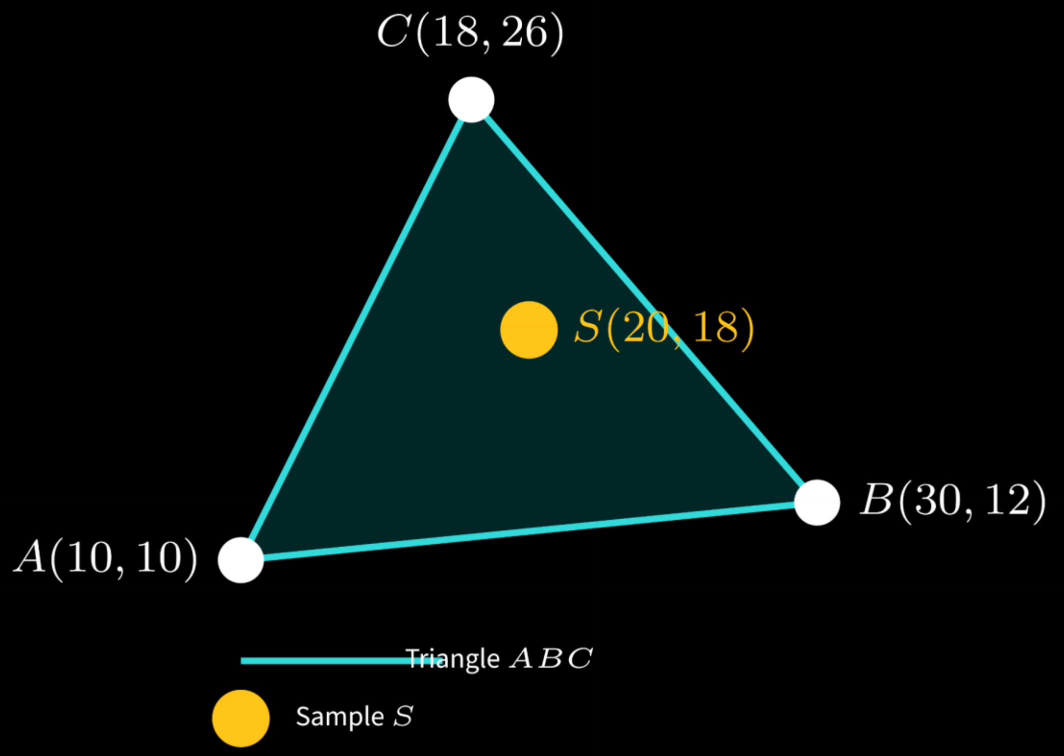

Figure: Simple triangle $\triangle ABC$ with sample S (inside the triangle).

Compute:

$$\begin{aligned}
E_{AB}(S) &= (10-12)\cdot 20 + (30-10)\cdot 18 + 10\cdot 12 - 30\cdot 10 \\
&= (-2)\cdot 20 + 20\cdot 18 + 120 - 300 \\
&= -40 + 360 - 180 = 140, \\
E_{BC}(S) &= (12-26)\cdot 20 + (18-30)\cdot 18 + 30\cdot 26 - 18\cdot 12 \\
&= (-14)\cdot 20 + (-12)\cdot 18 + 780 - 216 \\
&= -280 - 216 + 564 = 68, \\
E_{CA}(S) &= (26-10)\cdot 20 + (10-18)\cdot 18 + 18\cdot 10 - 10\cdot 26 \\
&= 16\cdot 20 + (-8)\cdot 18 + 180 - 260 \\
&= 320 - 144 - 80 = 96.
\end{aligned}$$

All positive $\Rightarrow$ *covered*. Normalize:

$$\lambda_a = \frac{68}{140+68+96} = \frac{68}{304} \approx 0.2237,\ \ \lambda_b = \frac{96}{304} \approx 0.3158,\ \ \lambda_c = \frac{140}{304} \approx 0.4605.$$

If vertex color $C_a{=}(1,0,0)$, $C_b{=}(0,1,0)$, $C_c{=}(0,0,1)$,

$$C(S) = 0.2237(1,0,0) + 0.3158(0,1,0) + 0.4605(0,0,1) \approx (0.224, 0.316, 0.461).$$

- Edge tests: integer-friendly, incremental across pixels, match hardware rules (top-left).
- Perspective-correct: divide attributes by w, interpolate, then re-multiply (avoid affine drift).
- Early-Z: if shader kills or writes depth unpredictably, early-Z may be disabled (perf hit).

Immediate vs. Deferred vs. Hybrid Pipelines

Immediate/forward shades during rasterization; *deferred* stores geometry attributes in a *G-buffer* then lights in a later pass; *hybrid* mixes raster with compute or ray tracing.

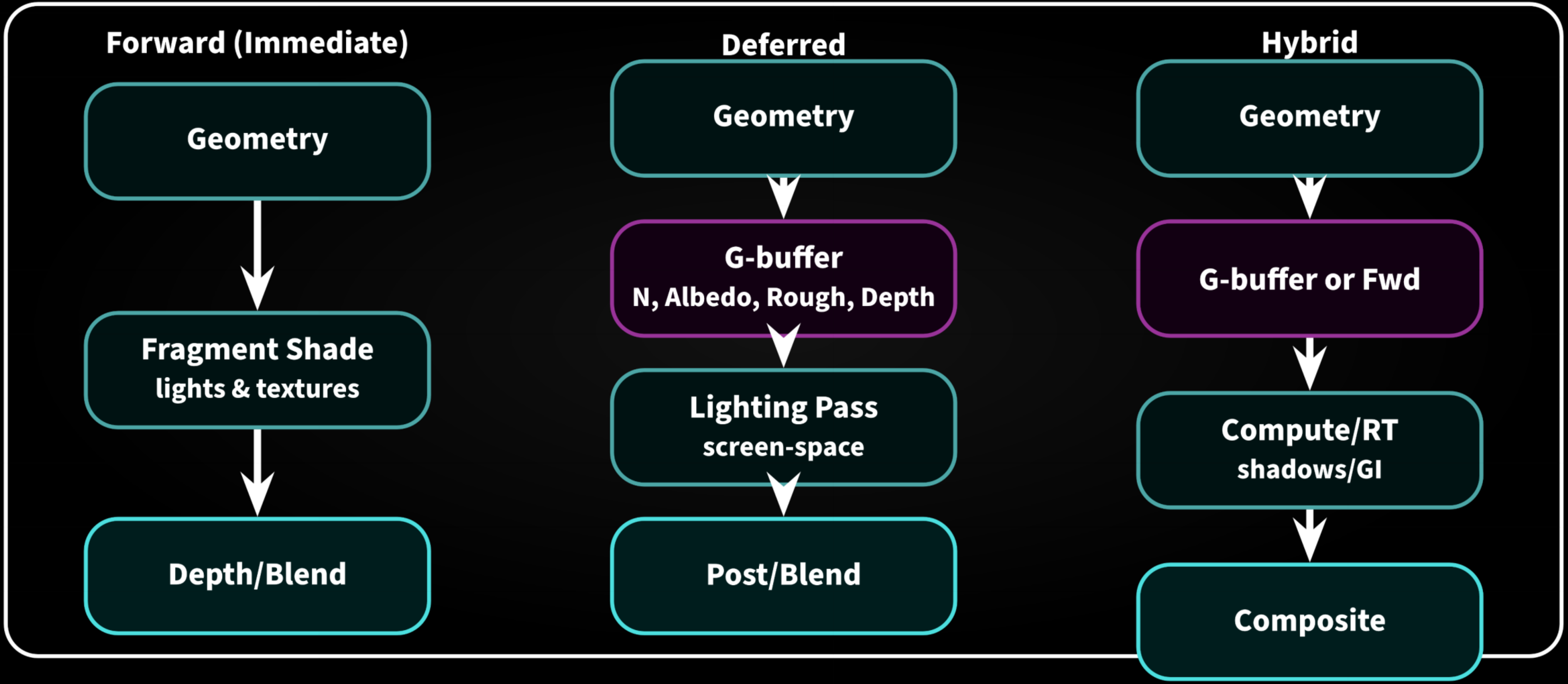

Figure 3: Forward shades during raster; Deferred writes a G-buffer then lights per-pixel; Hybrid adds compute/RT passes (shadows/GI).

Lambert + microfacet BRDF, deferred lighting

For a pixel with albedo a, normal $\mathbf{n}$, view $\mathbf{v}$, light L_i from direction $\mathbf{l}$:

$$f(\mathbf{l},\mathbf{v}) = \underbrace{\frac{a}{\pi}(\mathbf{n}\cdot\mathbf{l})_+}_{\text{Lambert}} + \underbrace{\frac{D(h)\,F(\mathbf{v}\cdot\mathbf{h})\,G(\mathbf{l},\mathbf{v})}{4(\mathbf{n}\cdot\mathbf{l})_+(\mathbf{n}\cdot\mathbf{v})_+}}_{\text{Cook–Torrance}}, \quad \mathbf{h} = \frac{\mathbf{l}+\mathbf{v}}{\|\mathbf{l}+\mathbf{v}\|}.$$

With GGX:

D(h)=$\alpha^2 \frac{}{\pi\left[(\mathbf{n}\cdot\mathbf{h})^2(\alpha^2-1)+1\right]^2,}$

F(x)=F$_0 + (1 - F_0)(1-x)^5,$

G(l,v)=2(n$\cdot\mathbf{v}) \frac{}{(\mathbf{n}\cdot\mathbf{v})+\sqrt{\alpha^2+(1-\alpha^2)(\mathbf{n}\cdot\mathbf{v})^2}\times(\mathbf{l}\leftrightarrow\mathbf{v}).}$

Single light, deferred

Given G-buffer: $a = (0.8, 0.2, 0.2)$, $\mathbf{n} = (0, 0, 1)$, roughness $r = 0.5$ ($\alpha = r^2 = 0.25$), $F_0 = 0.04$, camera at $\mathbf{v} = (0, 0, 1)$, light dir $\mathbf{l} = \frac{1}{\sqrt{3}}(1, 1, 1)$, intensity $I = 3$.

Lambert: $(\mathbf{n}\cdot\mathbf{l}) = 1/\sqrt{3} \approx 0.577$,

$$L_d = \frac{a}{\pi}(\mathbf{n}\cdot\mathbf{l})I \approx (0.8, 0.2, 0.2) \cdot \frac{0.577 \cdot 3}{\pi} \approx (0.441, 0.110, 0.110).$$

Specular: $\mathbf{h} = \frac{\mathbf{l}+\mathbf{v}}{\|\cdot\|} = \frac{(1,1,1)+(0,0,1)}{\|(1,1,2)\|} = \frac{(1,1,2)}{\sqrt{6}}$, $\mathbf{n}\cdot\mathbf{h} = 2/\sqrt{6} \approx 0.816.$

$$D = \frac{0.25^2}{\pi\,[0.816^2(0.25^2 - 1) + 1]^2} = \frac{0.0625}{\pi\,(0.666(-0.9375) + 1)^2} \approx \frac{0.0625}{\pi\,(0.375)^2} \approx 0.141.$$

$F(\mathbf{v}\cdot\mathbf{h})$ with $\mathbf{v}\cdot\mathbf{h} = 0.816$:

$$F = 0.04 + (1 - 0.04)(1 - 0.816)^5 \approx 0.04 + 0.96 \cdot (0.184)^5 \approx 0.040 + 0.0004 \approx 0.0404.$$

Smith G (symmetric) with $\mathbf{n}\cdot\mathbf{v} = 1, \mathbf{n}\cdot\mathbf{l} = 0.577$:

$$G \approx \frac{2}{1 + \sqrt{0.25^2 + (1 - 0.25^2) \cdot 1^2}} \cdot \frac{2 \cdot 0.577}{0.577 + \sqrt{0.25^2 + (1 - 0.25^2) \cdot 0.577^2}} \approx 0.97 \times 0.85 \approx 0.825.$$

Cook–Torrance term:

$$L_s = \frac{D\,F\,G}{4(\mathbf{n}\cdot\mathbf{l})(\mathbf{n}\cdot\mathbf{v})} I \approx \frac{0.141 \cdot 0.0404 \cdot 0.825}{4 \cdot 0.577 \cdot 1} \cdot 3 \approx 0.024.$$

Add RGB as energy-scalars: $L \approx L_d + L_s \approx (0.441, 0.110, 0.110) + (0.024, 0.024, 0.024) = (0.465, 0.134, 0.134)$.

- Forward: simple, great for few lights and MSAA; translucency easy; heavy overdraw can hurt.
- Deferred: lighting cost $\propto$ pixels, not geometry; memory bandwidth and translucency are challenges.
- Hybrid: keep fast raster base; offload hard effects (RT shadows/GI/SSR) to compute or RT stages.